

UDP-Glucose and UDP-Galactose Biosynthesis: A Three-Enzyme Bacterial Supply Module

A concise review-style synthesis. Citations are given as PubMed IDs (PMID). Claims are qualified where the evidence is indirect or organism-specific.

1. Executive summary

In bacteria, the nucleotide sugars UDP-glucose (UDP-Glc) and UDP-galactose (UDP-Gal) are produced by a compact, near-universal module of three enzymatic activities acting in series:

1. **Phosphoglucomutase** (Pgm, or the bifunctional AlgC/PMM-PGM class) interconverts glucose-6-phosphate (G6P) and glucose-1-phosphate (G1P).
2. **UTP-glucose-1-phosphate uridylyltransferase** (GalU / UDP-glucose pyrophosphorylase, UGP; EC 2.7.7.9) condenses G1P with UTP to give UDP-Glc + pyrophosphate (PPi).
3. **UDP-glucose 4-epimerase** (GalE; EC 5.1.3.2) reversibly epimerizes UDP-Glc at C4 to UDP-Gal.

The module is best understood as a **metabolic supply hub**: its single committed product, UDP-Glc, is the branch point from which UDP-Gal and a large set of downstream glycoconjugate and storage pathways draw activated hexose. The first two reactions are obligatory and sequential for *de novo* synthesis; the GalE reaction is a **reversible, accessory branch** that both generates UDP-Gal and connects to catabolic galactose salvage (the Leloir pathway).

Mechanistically the three steps are unusually well characterized. The mutase uses a phosphoserine "ping-pong" relay and a dramatic re-orientation of a bisphosphate intermediate (

P 15238632

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GalU is an oligomerization-dependent nucleotidyltransferase, typically tetrameric in bacteria

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P 39704542);

and GalE is a short-chain dehydrogenase/reductase (SDR) that uses a tightly bound NAD^+ to transiently oxidize, rotate, and re-reduce a 4-ketopyranose intermediate

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P 26162744
P 9708982).
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The major boundary confusions—bacterial glycogen (ADP-glucose/GlgC, **not** UDP-Glc) and the catabolic Leloir enzymes GalK/GalT—are cleared up in §2.

2. Definition and biological boundaries

2.1 What is inside the module

The module is defined by **product**, not by pathway membership: it comprises exactly the reactions that convert $\text{G6P} \rightarrow \text{G1P} \rightarrow \text{UDP-Glc} \rightleftharpoons \text{UDP-Gal}$. Concretely:

Step	Activity	EC	Representative players
1	Phosphoglucomutase	5.4.2.2	Pgm (dedicated); AlgC/SpgM/PgmG (bifunctional PMM/PGM)
2	UTP:G1P uridylyltransferase	2.7.7.9	GalU / UGP
3	UDP-glucose 4-epimerase	5.1.3.2	GalE

The **committed, module-defining** reaction is step 2 (GalU): phosphoglucomutase also serves glycolysis/gluconeogenesis and other nucleotide-sugar routes, so only when its G1P is drawn into GalU does the flux "belong" to this module.

2.2 Neighboring systems frequently confused with it (and why they are separate)

- **Bacterial glycogen synthesis is NOT part of this module.** In most bacteria and cyanobacteria the glucosyl donor for glycogen is **ADP-glucose**, made by glucose-1-phosphate *adenylyltransferase* **GlgC** (EC 2.7.7.27), an allosterically regulated enzyme

(activated by 3-PGA/FBP, inhibited by Pi) that shares G1P with GalU but diverges to a different nucleotide sugar

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Thus GalU and GlgC represent a **branch point at G1P**, not a shared pathway. (Plants/animals use UDP-Glc for storage glucans; bacteria generally do not—an important cross-kingdom caveat.)

- **The catabolic Leloir pathway (GalK → GalT → GalE) is a separate, neighboring system.** Galactokinase (GalK, a GHMP kinase;

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 and galactose-1-phosphate uridylyltransferase (GalT/GALT) salvage exogenous galactose. GalT runs a **double-displacement** mechanism through a covalent UMP–histidine intermediate, swapping UMP between Gal-1-P and UDP-Glc

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 —a chemistry unrelated to GalU's single-displacement nucleotidyl transfer. **GalE is the only enzyme genuinely shared** between the biosynthetic module and catabolic salvage. Equating "the Leloir pathway" with "UDP-Glc/UDP-Gal biosynthesis" is a common definitional error.

- **Downstream consuming pathways are outside the module.** UDP-Glc/UDP-Gal feed lipopolysaccharide (LPS) core/O-antigen, capsular polysaccharide (CPS), exopolysaccharide/biofilm matrix, wall teichoic acid (WTA) glycosylation, trehalose (OtsA uses UDP-Glc), osmoregulated periplasmic glucans, and glycolipids. These are *sinks*, not part of the supply module (see §6).

2.3 Competing definitions

Two definitional axes create most of the literature ambiguity: (i) "**Leloir pathway**" is often used loosely to include or exclude GalE and phosphoglucomutase; here we treat GalK/GalT as catabolic and outside scope. (ii) "**UGP**" is used for both the bacterial GalU family and the structurally distinct eukaryotic UDP-glucose pyrophosphorylases; the two are analogous, not orthologous (see §4.2, §5).

3. Mechanistic overview

3.1 Best current model of the sequence of events



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▼ (downstream sinks: LPS, CPS, EPS, WTA-Gal, ...)

- **Step 1 (mutase):** reversible; freely equilibrates G6P/G1P. Direction is pulled toward G1P when downstream steps consume it.
- **Step 2 (GalU):** the **committed, quasi-irreversible in vivo** step. Although chemically reversible, cytoplasmic **pyrophosphatase hydrolysis of PPi** pulls the reaction toward UDP-Glc, making this the physiological commitment point and the flux-controlling node.
- **Step 3 (GalE):** freely **reversible**; sets the UDP-Glc:UDP-Gal ratio according to demand. Because de novo cells make little free galactose, UDP-Gal is normally produced *from* UDP-Glc by GalE rather than by salvage.

3.2 Obligatory vs conditional vs accessory

- **Obligatory** for any UDP-Glc output: steps 1 and 2 (unless UDP-Glc is imported/salvaged, which bacteria generally cannot do at scale).
- **Accessory/branch:** step 3 (GalE) is required only when UDP-Gal is needed; it is dispensable for pure UDP-Glc supply.
- **Conditional players:** the phosphoglucomutase activity can be delivered by a dedicated Pgm **or** a bifunctional AlgC-type enzyme that also makes mannose-1-phosphate for GDP-mannose/alginate; which one is present is lineage- and niche-dependent (§4.1).

3.3 Chemistry of each step

- **Phosphoglucomutase (α -D-phosphohexomutase superfamily).** A conserved **catalytic phosphoserine** phosphorylates the substrate to a **glucose-1,6-bisphosphate** intermediate; the intermediate then **reorients** $\sim 180^\circ$ in the active site so the other phosphate can be transferred back to the enzyme, yielding product with the phosphate moved from C1 to C6 (or vice versa). A conserved **His (His329 in *P. aeruginosa* PMM/PGM), or an analogous Lys in the PGM subgroup**, acts as the general acid/base

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The phosphorylation state of the catalytic Ser strongly modulates enzyme flexibility and the required domain motions

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- **GalU (nucleotidyltransferase).** UTP attacks G1P; the α -phosphate of UTP is transferred to the phosphate of G1P forming the UDP-Glc pyrophosphate linkage with release of PPi. Single in-line displacement at phosphorus; Mg^{2+} -assisted. Broad 1-phosphosugar tolerance in vitro (e.g., *E. amylovora* GalU also uridylylates GlcNAc-1-P, xylose-1-P, mannose-1-P;

P 28844747).

- **GalE (SDR epimerase).** The tightly (essentially irreversibly) bound NAD^+ abstracts a hydride from C4 of the glucosyl/galactosyl moiety, generating a **4-ketopyranose** and NADH; the neutral intermediate **rotates within the active site** so that hydride is returned to the opposite face, inverting C4 stereochemistry with **no net redox change**

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Ser124/Tyr149 (*E. coli* numbering) form the catalytic couple.

4. Major molecular players and active assemblies

4.1 Phosphoglucomutase: two protein solutions to one activity

- **Dedicated Pgm** enzymes provide phosphoglucomutase activity as their primary role.
- **Bifunctional PMM/PGM (AlgC class).** AlgC (*P. aeruginosa*), SpgM (*Stenotrophomonas maltophilia*), and PgmG (*Spingomonas paucimobilis*) carry **both** phosphoglucomutase and phosphomannomutase activities in one four-domain polypeptide

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PgmG shows ~50-fold higher catalytic efficiency for G1P than M1P, i.e., it is "primarily" a PGM that can moonlight as a PMM

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These enzymes are the paradigm for the α -D-phosphohexomutase mechanism and are structurally the best-studied (*P. aeruginosa* PMM/PGM;

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Their **catalytic cycle demands large domain motions**, and loss of the bifunctional PGM reduces LPS and abolishes virulence in *S. maltophilia*

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—illustrating how one enzyme feeds several nucleotide-sugar modules.

4.2 GalU / UGP: oligomerization-dependent nucleotidyltransferase

Bacterial GalU adopts a Rossmann-like nucleotidyltransferase (dinucleotide-binding) fold; **activity is tied to oligomeric state, and most bacterial homologs are tetramers**

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Crystal structures from *Yersinia pestis*

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P 31475928
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and *Erwinia amylovora*

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reveal a conserved active site and support GalU as an **antibacterial/antivirulence drug target**—e.g., in silico-predicted GalU inhibitors block WTA galactosylation and attenuate *Listeria* virulence

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Note the important caveat that the **bacterial GalU family and the eukaryotic UGP family differ in oligomeric organization and are generally treated as distinct protein lineages performing the same reaction**. Bacterial GalU is typically tetrameric

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whereas the active eukaryotic enzyme is a **homo-octamer in human and yeast but**

essentially monomeric in most plants
([P 29494650](#))

—so even within eukaryotes the quaternary structure varies, and it differs again from bacteria. This is best read as convergence on a common chemistry rather than assumed strict orthology.

4.3 GalE: single-domain SDR homodimer with buried NAD⁺

GalE is a member of the **SDR superfamily** (Rossmann NAD-binding fold), usually a homodimer, with the cofactor bound so tightly it copurifies and is not exchanged during turnover

([P 26162744](#)).

Structures of *E. coli* GalE with UDP-Glc/UDP-Gal

([P 9708982](#))

and, more recently, cyanobacterial GalE (*Nostoc* all4713) in complex with UDP-Glc and NAD

([P 42250718](#))

define the substrate pocket and the geometry that permits ketopyranose rotation. **Substrate breadth differs by lineage:** single-domain bacterial GalE typically prefers UDP-hexoses, whereas some eukaryotic GALEs (human, *Drosophila*) additionally interconvert **UDP-GlcNAc/UDP-GalNAc**

([P 22654673](#)).

5. Evolutionary and cell-biological variation

- **Deep, modular ancestry from ancient folds.** Each step is built on an old scaffold: the α -D-phosphohexomutase superfamily (phosphoserine-relay mutases;

([P 15238632](#)),

the nucleotidyltransferase/Rossmann fold (GalU), and the NAD-dependent SDR/Rossmann fold (GalE). The **reaction logic is conserved from bacteria to humans**, but the specific proteins performing GalU differ in quaternary structure between prokaryotes and eukaryotes (bacterial tetramer vs eukaryotic octamer/monomer;

(P 39704542

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consistent with **convergent recruitment to the same chemistry** rather than assumed strict orthology.

- **Best representatives of the ancestral role.** For the mutase, the *P. aeruginosa* PMM/PGM is the mechanistic archetype, but the **dedicated Pgm** (mono-functional) likely reflects the ancestral "just PGM" role, with **bifunctional PMM/PGM an elaboration** that co-opted the same active site for mannose. For GalE, the **single-domain, UDP-hexose-specific bacterial enzyme** is the best model of the ancestral epimerase; the dual UDP-hexose/UDP-HexNAc activity of metazoan GALE is a later broadening.
- **Lineage-specific wiring.** Which phosphoglucomutase is used (Pgm vs AlgC-type) tracks with a lineage's glycan repertoire—alginate/gellan/EPS producers favor the bifunctional enzyme because it simultaneously supplies M1P. GalE presence/absence and specificity track with whether an organism builds galactose- or GalNAc-containing surface glycans.
- **Physiological-state variation.** Expression of galU/galE is frequently up-regulated under biofilm, stress, or host-colonization conditions; e.g., galU carbon flux is partitioned between exopolysaccharide synthesis and the pentose phosphate pathway under changing oxygen tension in *Komagataeibacter*

(P 32270472).

This is regulatory tuning of a shared hub rather than a change in the core chemistry.

- **Cross-kingdom caveat (do not overgeneralize).** Storage-glucan chemistry differs sharply: **bacteria = ADP-glucose/GlgC; plants/animals = UDP-glucose**. Galactose toxicity phenotypes and UPR involvement described in yeast and mammals

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reflect eukaryotic salvage physiology and should not be transposed onto the bacterial biosynthetic module.

6. Constraints, dependencies, and failure modes

- **Order is fixed for de novo synthesis:** G6P must become G1P before GalU can act, and UDP-Glc must exist before GalE can make UDP-Gal. There is no bacterial route that makes UDP-Gal from UDP-Glc without GalE, nor UDP-Glc from G1P without a nucleotidyltransferase.
- **Thermodynamic pull:** GalU is chemically reversible but rendered effectively forward in vivo by **PPi hydrolysis**; without pyrophosphatase activity the commitment step would stall. This is the key "directional" constraint.
- **Cofactor dependency:** GalE is inert without its bound NAD⁺; the cofactor is regenerated intramolecularly each cycle, so net turnover consumes no NAD(H)—but apo-enzyme or reduced/adducted NAD states are inactive.
- **Substrate-specific branching at G1P:** G1P is contested by GalU (→UDP-Glc) and GlgC (→ADP-Glc); these are **mutually exclusive fates** for a given G1P molecule and are separately regulated (allosteric control of GlgC;

P 24112771

P 39639537

Mannose branches away earlier (M1P via PMM/AlgC).

- **Failure modes (loss-of-function phenotypes, all in downstream sinks):** galU loss → truncated LPS core, colonization and biofilm defects, heightened sensitivity to bile/complement/antimicrobials (*Vibrio cholerae*,

P 11119535

Xanthomonas,

P 25033927

galE loss → inability to make UDP-Gal-containing surface glycans and loss of galactose catabolism, but often near-normal LPS if galactose is not a core sugar

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These phenotypes belong to consuming pathways, confirming the module's role as **supplier**, not effector.

7. Controversies and open questions

1. **In vivo directionality and flux control.** The commitment step is generally assigned to GalU (PPi-pulled), but quantitative flux control coefficients across the mutase/GalU/GalE trio are rarely measured; the relative control under biofilm vs planktonic states is largely inferred, not directly demonstrated.
2. **Oligomeric state and regulation of GalU.** The tetramer is the consensus (P 39704542), but whether oligomerization is a *regulatory* switch (allostery, UTP/UDP-Glc feedback) or merely a stability requirement is unsettled and likely species-specific.
3. **"Which PGM?" ambiguity.** Many genomes encode multiple phosphohexomutases (dedicated Pgm, AlgC-type, phosphoglucosamine mutase GlmM/related; note substrate-assisted catalysis in that subfamily, P 29967067). Assigning the physiologically dominant supplier of G1P to a specific gene is often done by homology rather than by flux measurement.
4. **GalE substrate scope generalizations.** The UDP-HexNAc epimerase activity is prominent in metazoan GALE (P 22654673) and some bacteria but absent in others; reviews sometimes overstate promiscuity by pooling data across distant organisms (P 26162744). Substrate range should be asserted per-enzyme, not per-family.
5. **Evolutionary origin of the bacterial vs eukaryotic UGP split.** The bacterial and eukaryotic enzymes differ in quaternary structure (tetramer vs octamer/monomer; P 39704542); P 29494650); whether this reflects deep divergence within one fold family or independent recruitment is not fully resolved and merits explicit phylogenetic caution.
6. **Cross-organism transfer of mechanism.** Much galactosemia/UPR mechanism is eukaryotic

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P 35301088
P 21793161);
 its relevance to bacterial physiology is limited and should not be assumed.

8. Key references

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P 32270472

Explicit uncertainty statement

This synthesis draws on structural and enzymological work concentrated in a handful of model organisms (*P. aeruginosa* PMM/PGM; *E. coli*/*Nostoc* GalE; *Y. pestis*/*E. amylovora* GalU). Mechanistic chemistry (phosphoserine relay, PPI-pulled nucleotidyl transfer, NAD⁺-mediated C4 epimerization) is strongly supported and broadly transferable. In contrast, regulation, oligomeric-state control, quantitative flux control, and per-species substrate breadth are less uniformly established and should not be generalized across all bacteria without direct evidence.